CITIROC STUDY

PREAMPLIFIER
FAST SHAPER
SLOW SHAPER

Y. FAVRE

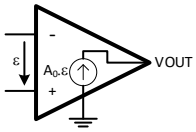
08.01.2025

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1 NON IDEAL OPA TRANSFER FUNCTION

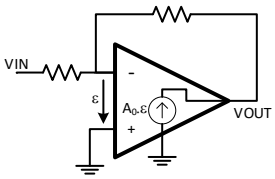
1.1 OPA open loop transfer function



Open loop gain bandwidth: $A_V = \frac{A_0}{1+\frac{p}{w_0}}$, $w_0 = 2\pi f_0$

$V_0 = A_V.\varepsilon = A_V.(V_+ - V_-)$

1.2 Close loop inverter OPA



$$V_o = A_V.\varepsilon \, , V_- = \frac{Z_1.V_0+Z_2.V_{IN}}{Z_1+Z_2}, \varepsilon = -\frac{Z_1.V_0+Z_2.V_{IN}}{Z_1+Z_2}$$

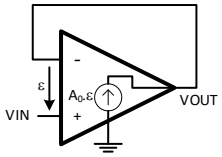
$$\frac{V_o}{V_{IN}} = -\frac{Z_2.A_V}{Z_1.(1+A_V)+Z_2} = -\frac{Z_2.\frac{A_0}{1+\frac{p}{w_0}}}{Z_1.\left(1+\frac{A_0}{1+\frac{p}{w_0}}\right)+Z_2} = -\frac{Z_2.A_0}{Z_1.\left(1+\frac{p}{w_0}+A_0\right)+Z_2.(1+\frac{p}{w_0})} = -\frac{Z_2.A_0}{Z_1.(1+A_0)+Z_2}.\frac{1}{1+\frac{(Z_1+Z_2).p}{(Z_1.(1+A_0)+Z_2).w_0}}$$

$$\mathbf{TF(p)} = \frac{\mathbf{V_o}}{\mathbf{V_{IN}}} = -\mathbf{G_0}.\frac{\mathbf{1}}{\mathbf{1}+\frac{\mathbf{p}}{\mathbf{w_c}}}$$ = 1st order Low Pass Filter

DC Gain $\boldsymbol{G_0} = \frac{A_0}{\frac{Z_1}{Z_2}.(A_0+1)+1} \cong \frac{Z_2}{Z_1}$ for ($\frac{A_0.Z_1}{Z_2}$>>1)

Closed loop bandwidth $\mathbf{f_c} = \mathbf{f_0}.\frac{\mathbf{Z_1}.(1+A_0)+\mathbf{Z_2}}{\mathbf{Z_1}+\mathbf{Z_2}} \cong \mathbf{f_0}.\frac{\mathbf{Z_1}.A_0+\mathbf{Z_2}}{\mathbf{Z_1}+\mathbf{Z_2}} \cong \mathbf{f_0}.\frac{\mathbf{Z_1}.A_0}{\mathbf{Z_1}+\mathbf{Z_2}}$ for $Z_1.A_0 >> Z_2$

1.3 Closed loop follower OPA



$V_o = A_V.\varepsilon\,, V_- = V_0, \varepsilon = V_{IN} - V_0$

$$TF(p) = \frac{V_o}{V_{IN}} = \frac{A_V}{1+A_V} = \frac{\frac{A_0}{1+\frac{p}{w_0}}}{1+\frac{A_0}{1+\frac{p}{w_0}}} = \frac{A_0}{1+\frac{p}{w_0}+A_0} = \frac{A_0}{1+A_0}.\frac{1}{1+\frac{p}{(1+A_0).w_0}} \cong \frac{\mathbf{1}}{\mathbf{1}+\frac{p}{\mathbf{(1+A_0).w_0}}}$$ with large A_0.

DC Gain =1, Closed loop bandwidth $\mathbf{f_C = (1+A_0).f_0}$

1.4 CITIROC parameters

OPA	A_0 (dB)	A_0	f_0 (kHz)	f_C (MHz)
HG	65	1778	450	800
LG	58	794	166	132
FS	60	1000	1200	1200
SS	52	398	630	251

2 HG/LG PEAMPLIFIER

2.1 Theory

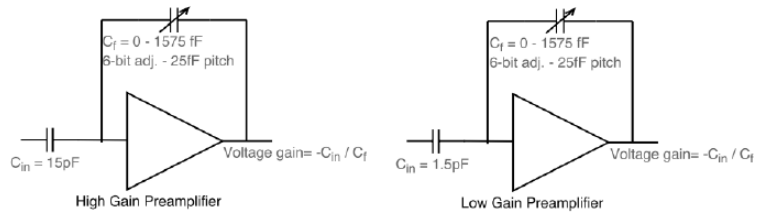
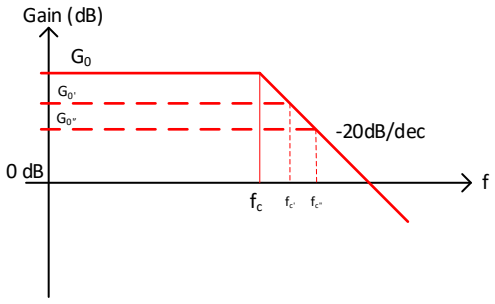


Figure 8 – High gain and low gain voltage sensitive preamplifier

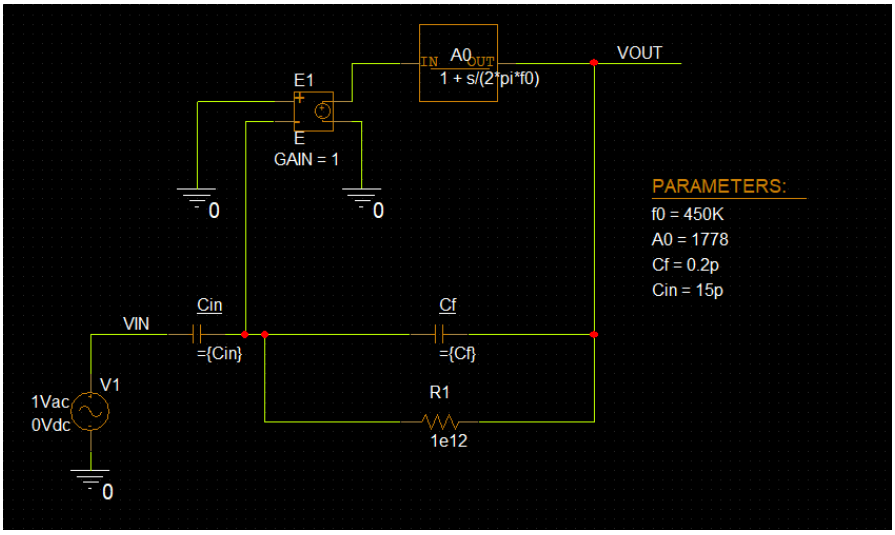
$Z_1 = \frac{1}{C_{IN}.p}$ $Z_2 = \frac{1}{C_f.p}$

$\frac{V_o}{V_{IN}} = -G_0.\frac{1}{1+\frac{p}{w_c}}$, $G_0 = \frac{A_0}{\frac{C_f}{C_{IN}}.(A_0+1)+1}$, Closed loop bandwidth $f_c = f_0.\frac{C_f.(1+A_0)+C_{IN}}{C_f+C_{IN}} \cong f_0.\frac{C_f.A_0}{C_f+C_{IN}}$

OPA	A_0 (dB)	A_0	f_0 (KHz)	C_{IN} (pF)	S/C gain	C_f (pF)	G_0	G_0 (dB)	f_c (MHz)
HG	65	1778	450	15	62	0.025	448.4	53.0	1.8
					55	0.2	71.9	37.1	11.0
					48	0.375	39.1	31.8	20.0
					31	0.8	18.5	25.4	41.0
					20	1.075	13.8	22.8	54.0
					0	1.575	9.5	21.3	76.5
LG	58	794	166	1.5	62	0.025	55.7	34.9	2.3
					55	0.2	7.4	17.4	15.7
					48	0.375	4.0	12.0	26.5
					31	0.8	1.9	5.4	46.0
					20	1.075	1.4	2.9	55.2
					0	1.575	1.0	-0.4	67.7

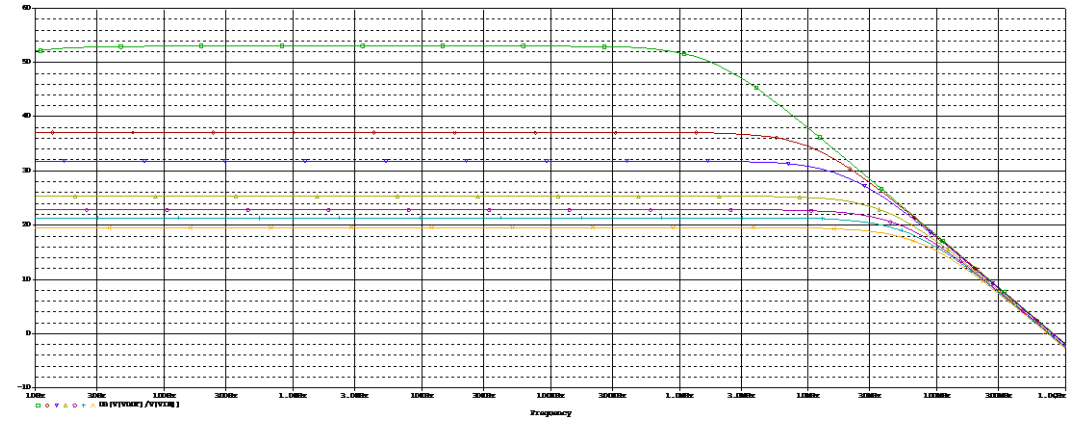


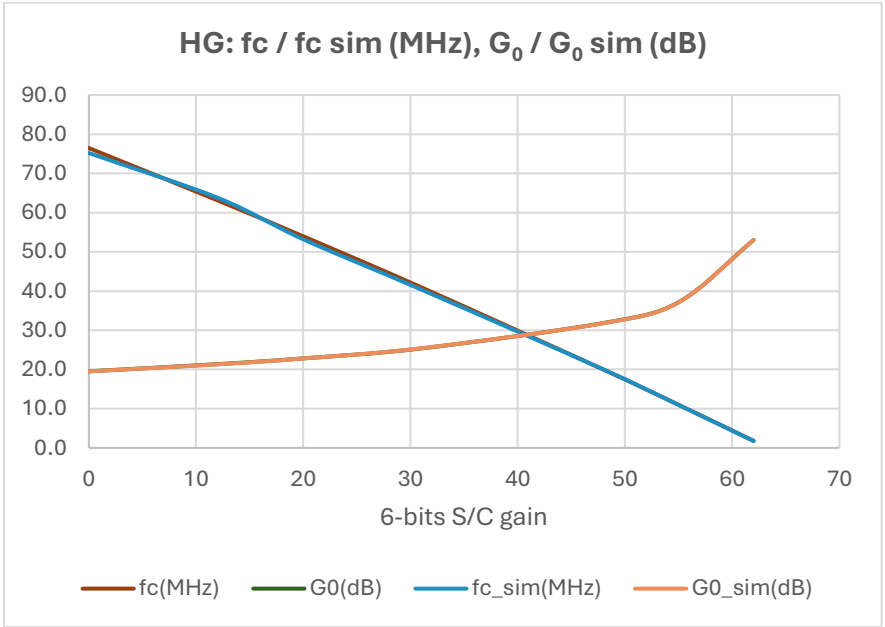
2.2 AC Simulation



Cf(F)= 0.025p 0.2p 0.375p 0.8p 1.075p 1.275p 1.575p

HG= 62 55 48 31 20 12 0





⇨ Excellent corelation between simulation and theory

3 FAST SHAPER

3.1 Theory

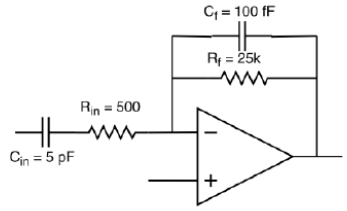


Figure 3 - Fast shaper block scheme

$Z_1 = \frac{1+R_{IN}.C_{IN}.p}{C_{IN}.p}$ $Z_2 = \frac{R_f}{1+R_f.C_f.p}$ $\tau = R_f.C_f = R_{IN}.C_{IN} = \frac{1}{w_{fs}}$

$$TF(p) = -\frac{Z_2.A_0}{Z_1.\left(1+\frac{p}{w_0}+A_0\right)+Z_2.\left(1+\frac{p}{w_0}\right)} \cong -\frac{A_0.R_f.C_{IN}.p}{(1+\tau.p)^2.\left(\frac{p}{w_0}+A_0\right)+R_f.C_{IN}.p.\left(1+\frac{p}{w_0}\right)} =$$

$$\frac{A_0.R_f.C_{IN}.p}{A_0+\left(2.A_0.\tau+\frac{1}{w_0}+R_f.C_{IN}\right).p+\left(A_0.\tau^2+\frac{2.\tau+R_f.C_{IN}}{w_0}\right).p^2+\frac{\tau^2}{w_0}.p^3} = \frac{R_f.C_{IN}.p}{1+(2.\tau+\frac{1}{A_0.w_0}+\frac{R_f.C_{IN}}{A_0.}).p+(\tau^2+\frac{2.\tau}{A_0.w_0}+\frac{R_f.C_{IN}}{A_0.w_0}).p^2+\frac{\tau^2}{A_0.w_0}.p^3}$$

1st approximation (simple):

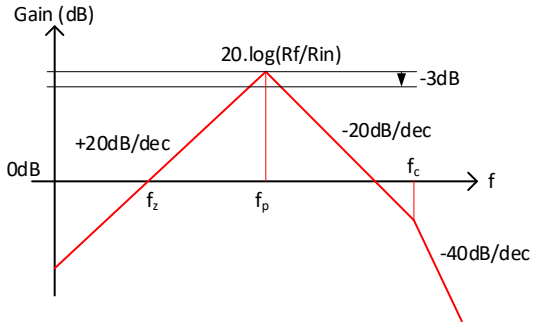
$TF(p) \cong \frac{R_f.C_{IN}.p}{(1+2\tau.p)^2.(1+\frac{p}{4.A_0.w_0})}$. This is 1st order approximation pass band filter centred around f_p (0dB at f_z, +20dB/dec up to f_p then -20dB/dec up to f_c and then -40dB/dec) with a peak gain of G_{fc}

$f_p = \frac{1}{2.\pi.2\tau} = \frac{1}{4.\pi\ R_{IN}.C_{IN}}$, $f_c = \frac{4.A_0.w_0}{2.\pi}$, $f_z = \frac{1}{2.\pi\ R_f.C_{IN}}$, $G_{fc} \cong \frac{R_f}{4.R_{IN}}$

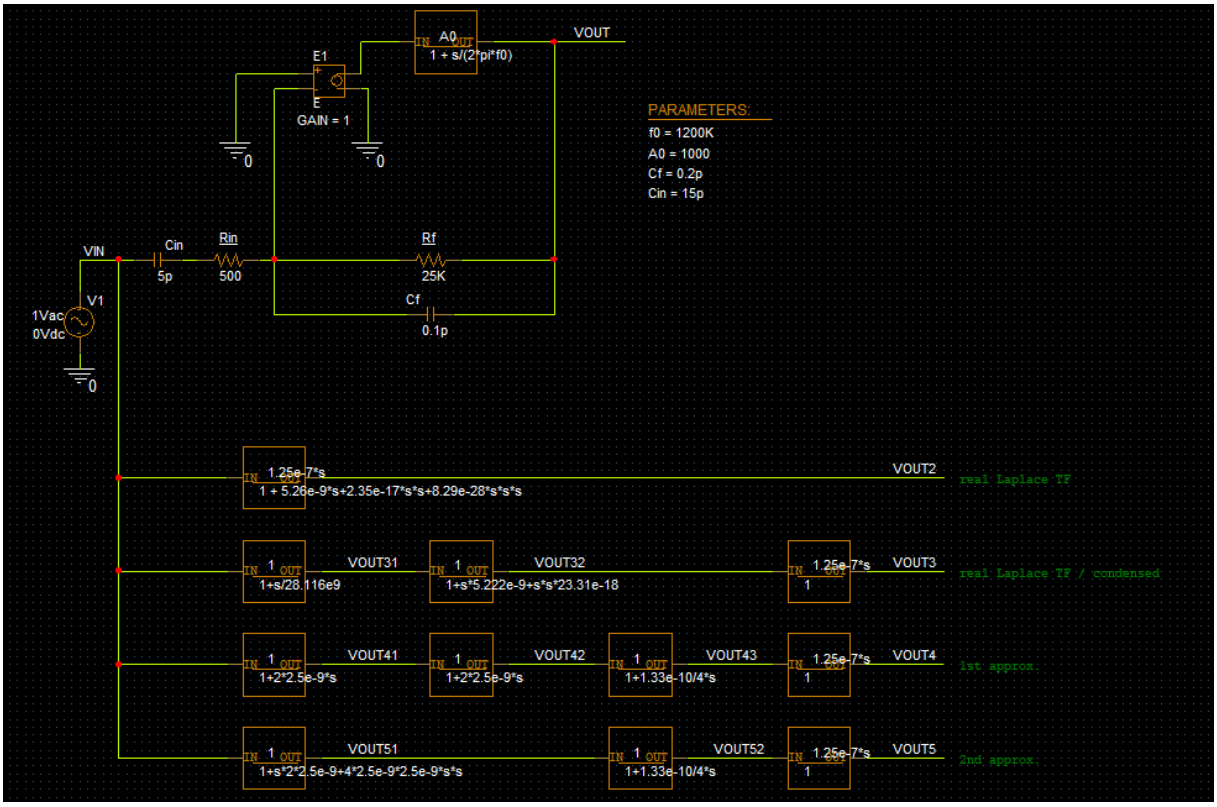
2nd approximation (better):

$TF(p) \cong \frac{R_f.C_{IN}.p}{(1+2\tau.p+4\tau^2p^2).(1+\frac{p}{4.A_0.w_0})}$. This is better approximation of the pass band filter having a 2nd order LPF dominant ($\frac{1}{1+\frac{2z}{w_n}p+\frac{p^2}{w_n{}^2}}$), followed by a 1st order LPF (high frequency pole that can be omitted). The 2nd order nominal frequency is $f_n = \frac{1}{2.\pi.2\tau} = \frac{1}{4.\pi\ R_{IN}.C_{IN}}$, z=0.5, resonant frequency $f_r = f_n.\sqrt{1-2.z^2}$, $G_{fn} = \frac{R_f}{2.R_i.2.z}$

Parameters		Full equation		1st Approximation		2nd Approximation	
Ri (Ω)	500	$2.\tau+\frac{1}{A_0.w_0}+\frac{R_f.C_{IN}}{A_0.}$	5.26E-09	$4\tau+\frac{1}{4.A_0.w_0}$	1E-08	$2\tau+\frac{1}{4.A_0.w_0}$	5.03E-09
Ci (pF)	5	$\tau^2+\frac{2.\tau}{A_0.w_0}+\frac{R_f.C_{IN}}{A_0.w_0}$	2.35E-17	$4\tau^2+\frac{\tau}{A_0.w_0}$	2.53E-17	$4\tau^2+\frac{\tau}{2.A_0.w_0}$	2.35E-7
Rf (kΩ)	25	$\frac{\tau^2}{A_0.w_0}$	8.29E-28	$\frac{\tau^2}{A_0.w_0}$	8.29E-28	$\frac{\tau^2}{A_0.w_0}$	8.29E-28
Cf (pF)	0.1	$R_f.C_{IN}$	1.25E-7			z	0.5
A0	1000	f_p (MHz)	31.8	f_p (MHz)	31.8	f_n (MHz)	31.8
f0 (MHz)	1.2	f_c (MHz)	4800	f_c (MHz)	4800	f_c (MHz)	4800
		f_z (MHz)	1.27	f_z (MHz)	1.27	f_z (MHz)	1.27
		G_{fc}	27.5	G_{fc}	21.9	G_{fn}	28

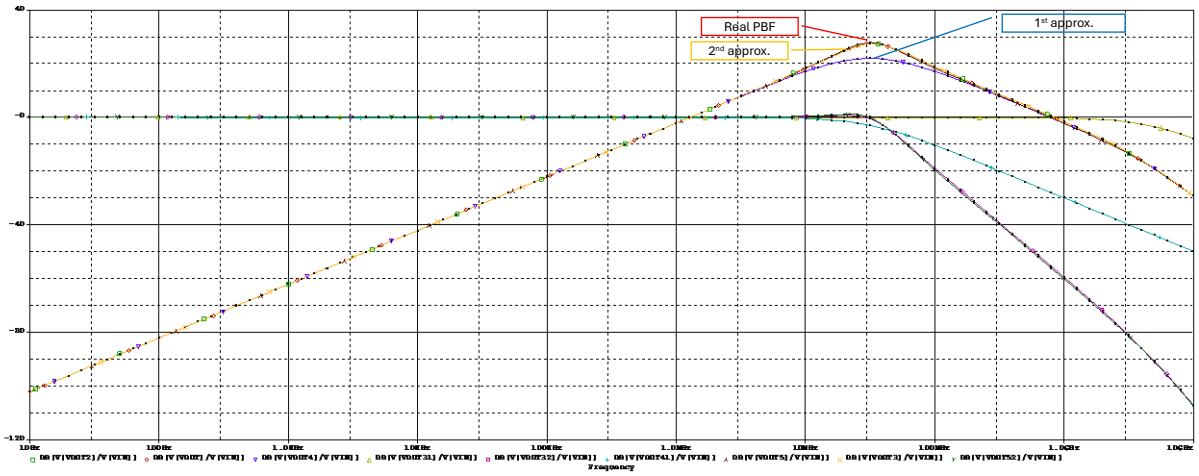


3.2 AC Simulation



VOUT = VOUT2 = VOUT3 ~=VOUT5

VOUT5 is decomposed into the 2nd order LPF (VOUT51, frequency f_n=31.8MHz, z=0.5) followed by the high frequency 1st order LPF (VOUT52, fc=4.8GHz)) and the 1st order zero (fz=1.27MHz).



The 1st approx.. and the 2nd order approx. have the same center frequency but the 2nd approx. is better with the gain.

4 PREAMPLIFIER + FAST SHAPER

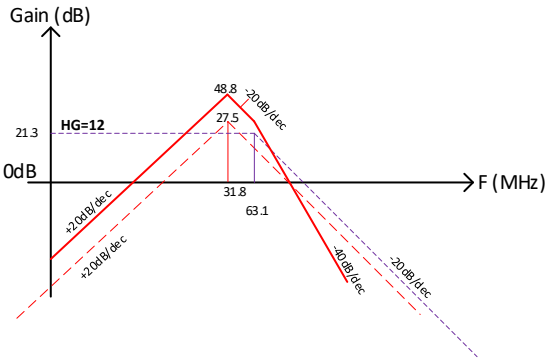
4.1 Theory

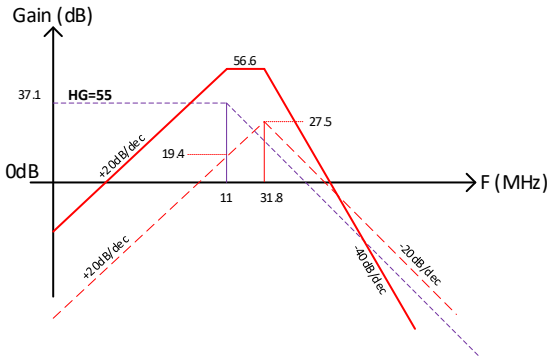
The transfer function is the convolution of both stages i.e. a multiplication in the Laplace domain with the corresponding A_0 and w_0 of each stage (fs = fast shaper, pa = pre-amplifier)

$$TF(p) \cong \frac{G_{0pa}.R_f.C_{IN}.p}{(1+2.\tau.p+4.\tau^2.p^2).\left(1+\frac{p}{4.A_{0fs}.w_{0fs}}\right).(1+\frac{p}{A_{0pa}.w_{0pa}})} \cong \frac{G_{0pa}.R_f.C_{IN}.p}{(1+2.\tau.p+4.\tau^2.p^2).(1+\frac{p}{A_{0pa}.w_{0pa}})}$$

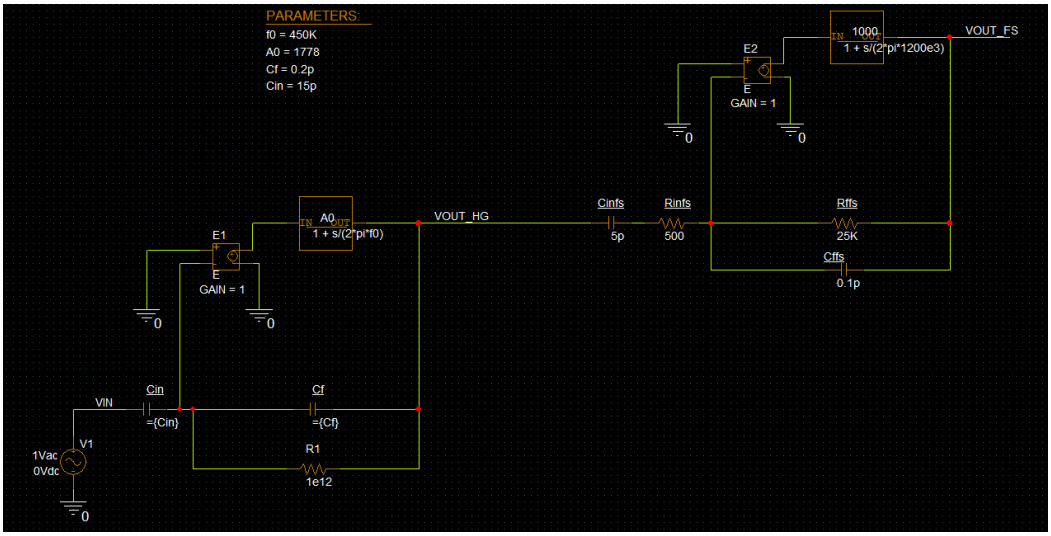
Depending on the PA gain, the cut-off frequency of the fast shaper is affected, the closed loop cutoff frequency of the fast shaper is too high and can be neglected.

OPA	S/C gain	C_f (pF)	G_{0pa} (dB)	f_{cpa} (MHz)	G_{fcfs}(dB)	f_{cfs} (MHz)
HG	62	0.025	53.0	1.8	27.5	31.8
	55	0.2	**37.1**	**11.0**		
	48	0.375	31.8	20.0		
	31	0.8	25.4	41.0		
	20	1.075	22.8	54.0		
	12	1.275	**21.3**	**63.1**		
	0	1.575	19.5	76.5		



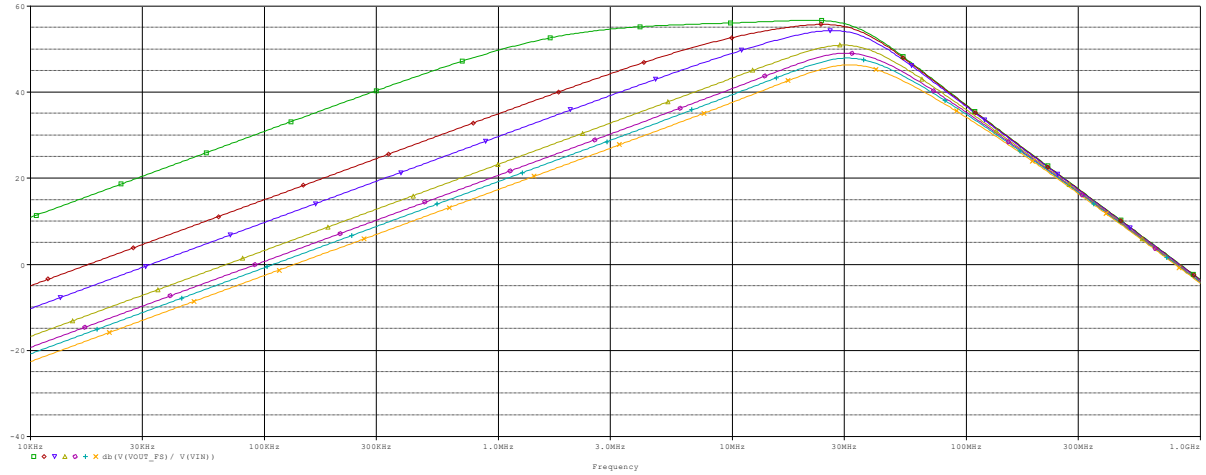


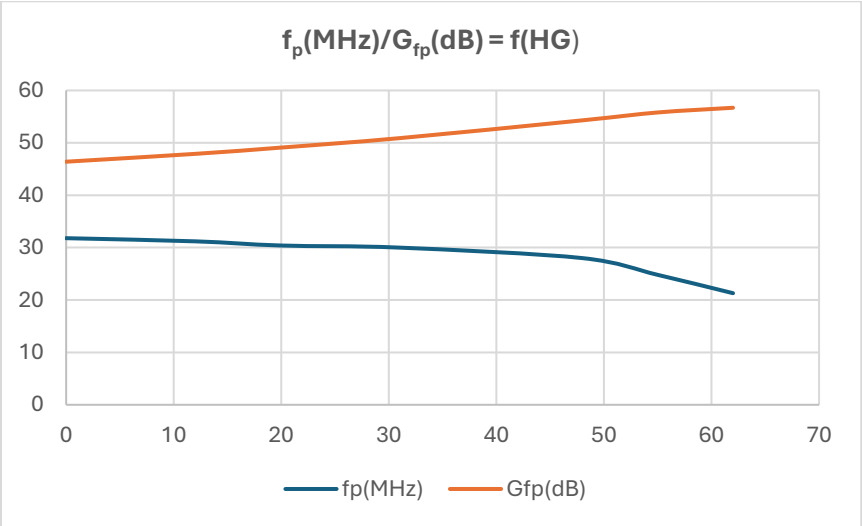
4.2 AC Simulation



C_f(F)= 0.025p 0.2p 0.375p 0.8p 1.075p 1.275p 1.575p

HG= 62 55 48 31 20 12 0

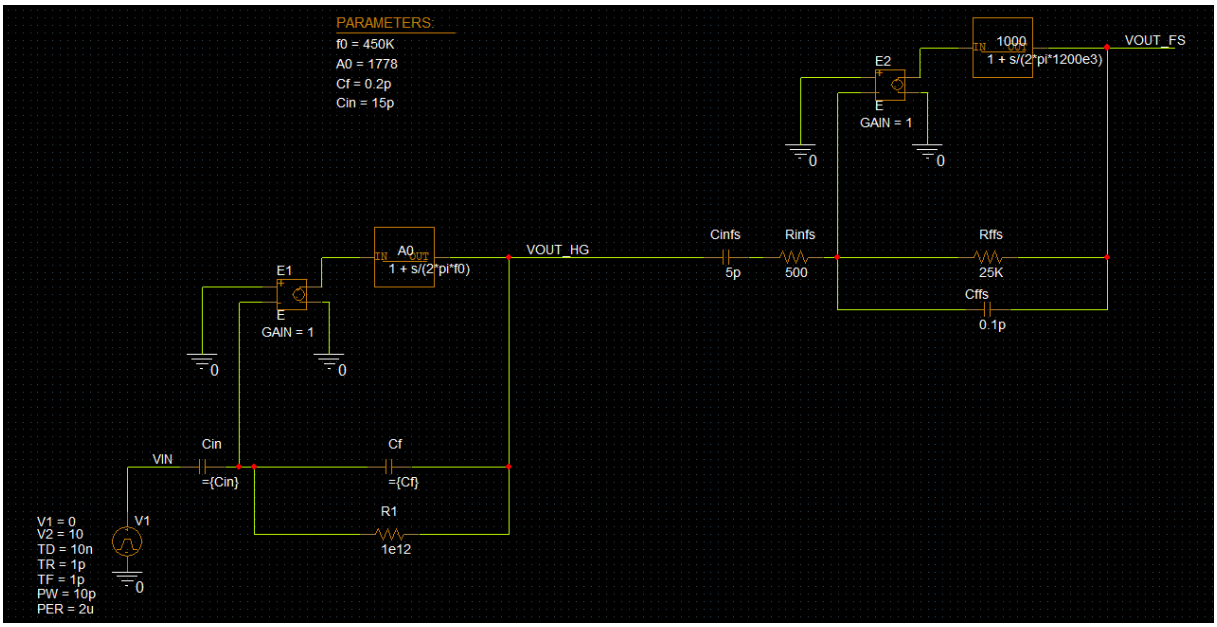




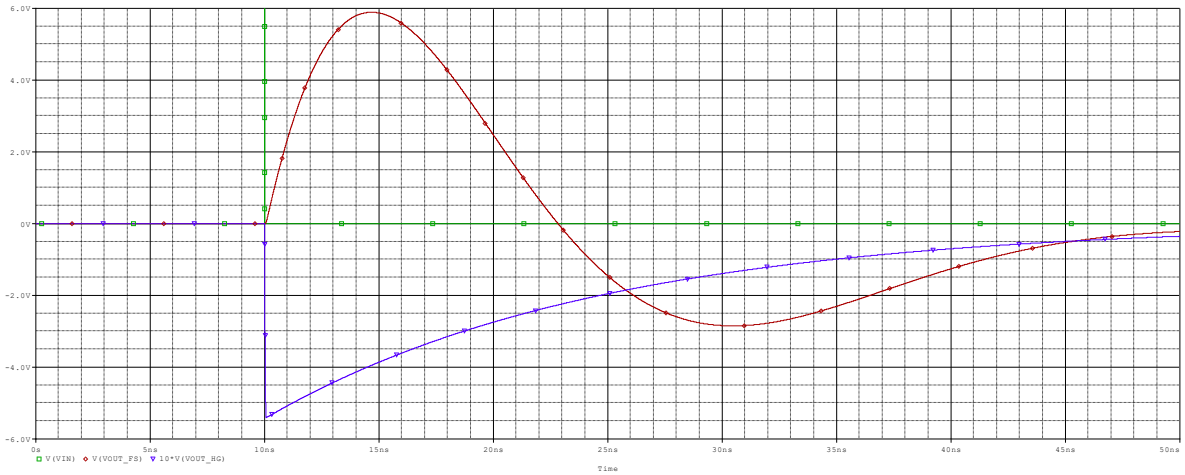
C_f(pF)	HG	f_p(MHz)	G_{fp}(dB)
0.025	62	21.3	56.7
0.2	55	24.8	55.8
0.375	48	28	54.3
0.8	31	30	50.9
1.075	20	30.4	49.1
1.275	12	31.2	47.9
1.575	0	31.8	46.4

4.3 Transient simulation

4.3.1 Dirac impulse input



Transient impulse response (dirac) with HG=55 (C_f=0.2pF):

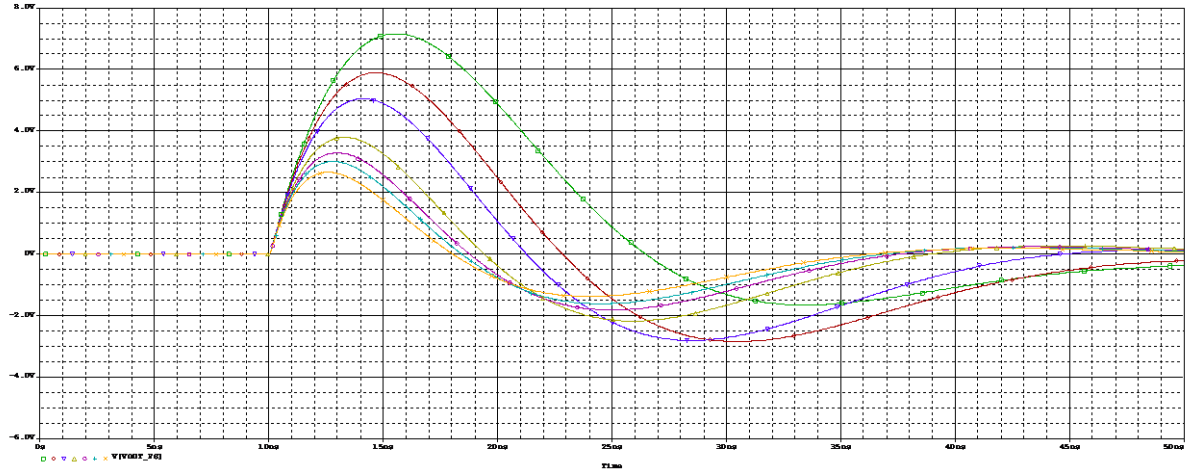


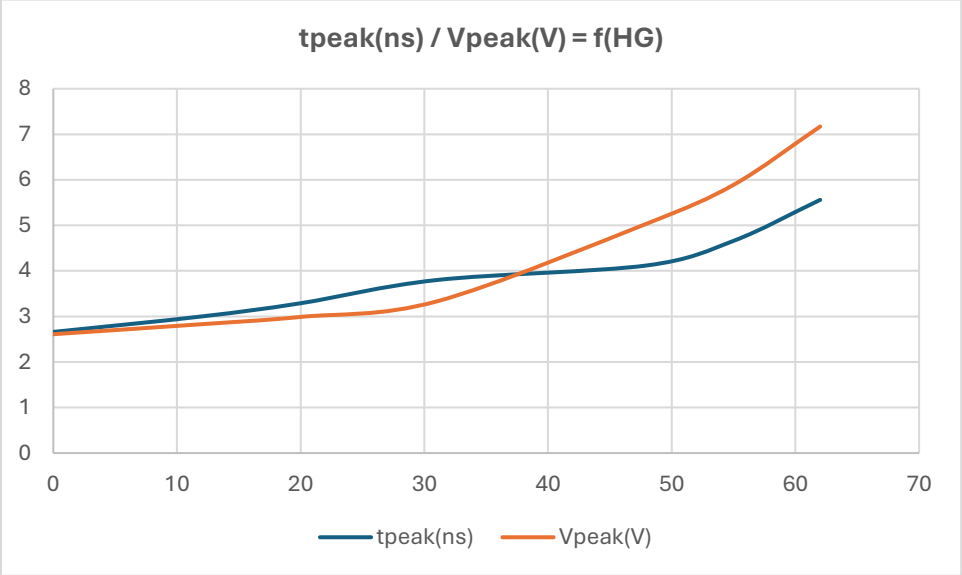
t_{peak} = 4.66ns, V_{peak} = 5.89V

NB: Dirac model = Pulse: tr=tf=1ps, pw=10ps, V=10V

C_f(F)= 0.025p 0.2p 0.375p 0.8p 1.075p 1.275p 1.575p

HG= 62 55 48 31 20 12 0

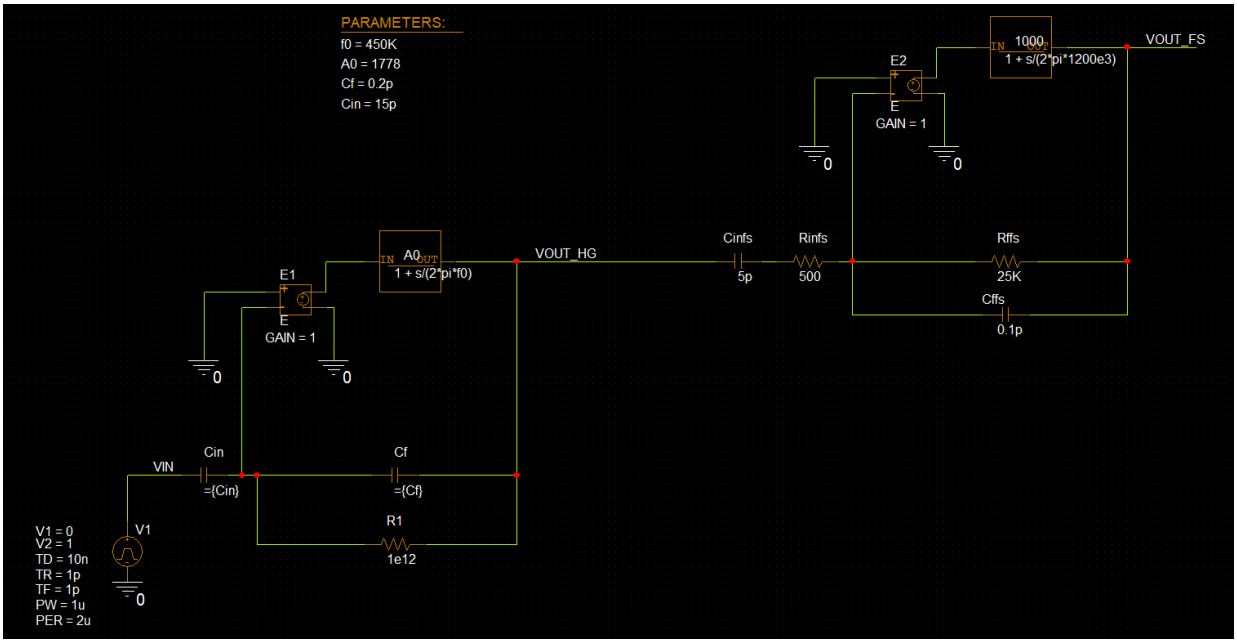




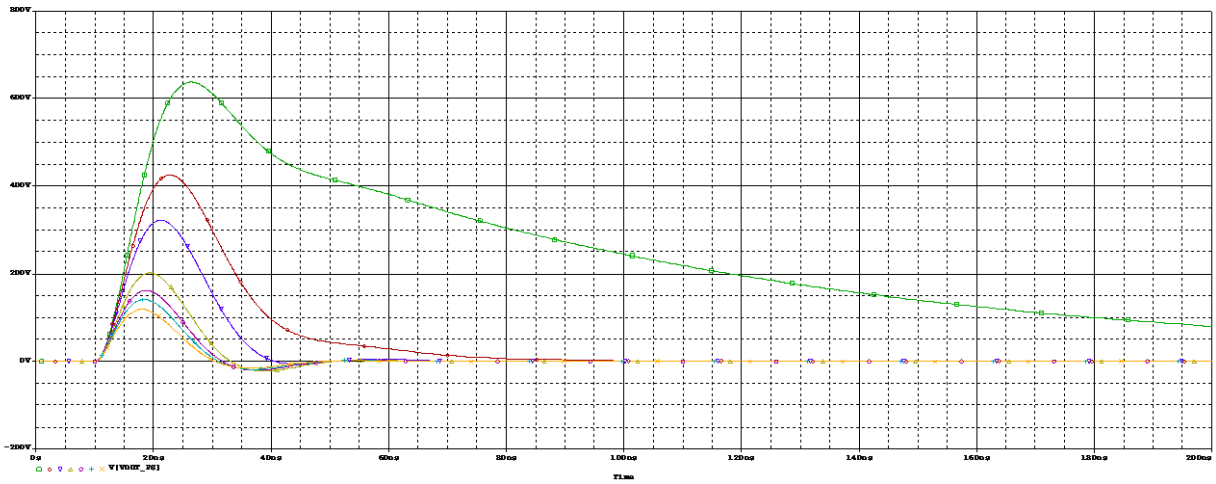
C_f(pF)	HG	t_{peak}(ns)	V_{peak}(V)
0.025	62	5.56	7.17
0.2	55	4.66	5.89
0.375	48	4.12	5.04
0.8	31	3.8	3.33
1.075	20	3.29	2.99
1.275	12	3	2.83
1.575	0	2.66	2.61

NB: 2.Rin.Cin Fast Shaper = 5ns

4.3.2 Step input

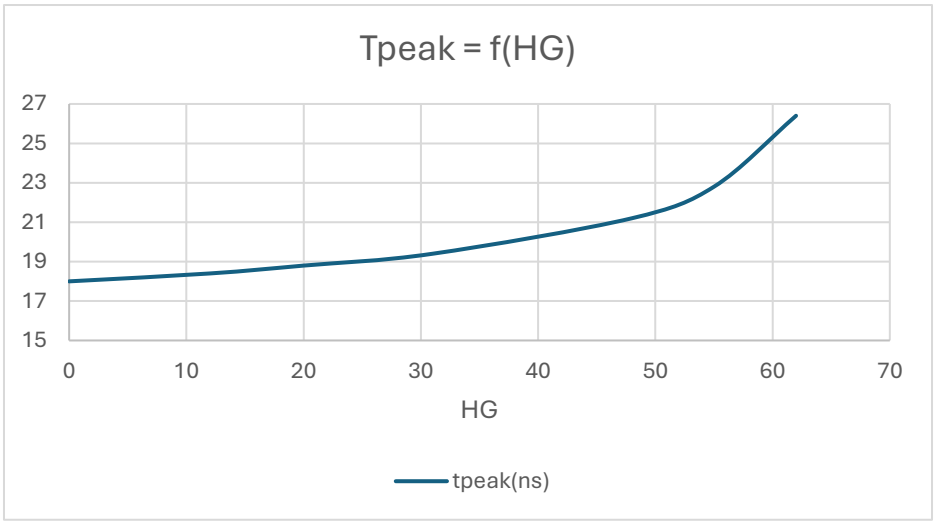


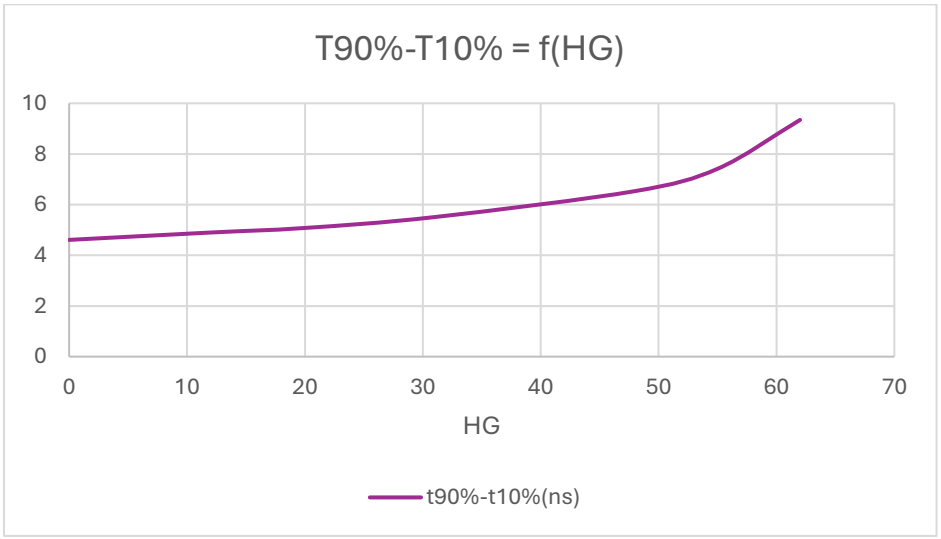
C_f(F)= 0.025p 0.2p 0.375p 0.8p 1.075p 1.275p 1.575p
HG= 62 55 48 31 20 12 0

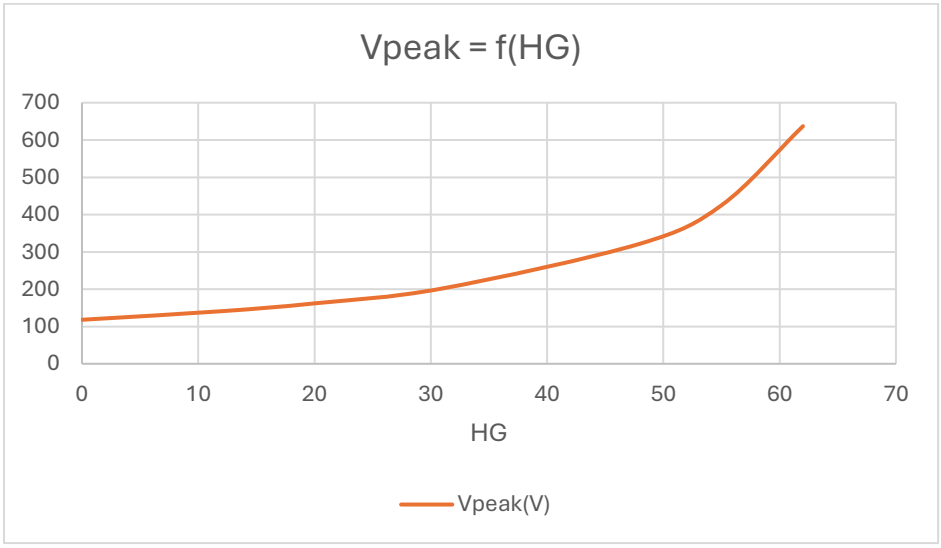


⇨ The slower queue on HG>=55 is due to the frequency plateau (allows more low frequency to pass through the filter) between f_{cpa} and f_{cfs} ($f_{cpa} < f_{cfs}$) which is not the case when HG<55 as seen on bode plots.

Cf(pF)	HG	tpeak(ns)	Vpeak(V)	t90%(ns)	t10%(ns)	t90%-t10%
0.025	62	26.4	637	21.875	12.53	9.345
0.2	55	22.8	425	19.515	12.1	7.415
0.375	48	21.2	322	18.395	11.86	6.535
0.8	31	19.4	202	17.01	11.5	5.51
1.075	20	18.8	162	16.48	11.4	5.08
1.275	12	18.4	141	16.23	11.33	4.9
1.575	0	18	118	15.84	11.23	4.61







⇨ **The preamplifier cutoff frequency plays a significant role on the peaking time as seen on bode plots. $T_{90\%-10\%}$ should be close to 2τ (5 ns) but with a shift again due to the PA.**

5 SLOW SHAPER

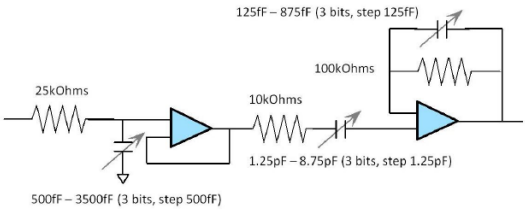


Figure 10 – Slow shaper schematic

Slow control value	Peaking time value
6 (110)	12.5 ns
5 (101)	25.0 ns
4 (100)	37.5 ns
3 (011)	50.0 ns
2 (010)	62.5 ns
1 (001)	75.0 ns
0 (000)	87.5 ns

All the RC couples have the same time constant τ=RC

The **1st stage $TF_1(p)$** is a 1st order RC low pass filter with a 1st order LPF non ideal follower:

$$TF_{1SS}(p) \cong \frac{1}{(1+\tau.p).(1+\frac{p}{A_0.w_0})}, \quad f_{c2} = \frac{A_0.w_0}{2.\pi} = 251MHz, f_{c1} = \frac{1}{2.\pi.\tau} = \frac{1}{2.\pi.RC}$$

OPA	A_0 (dB)	A_0	f_0 (KHz)	f_{c2} (MHz)	S/C	C (pF)	f_{c1} (MHz)
SS	52	398	630	251	0	0.5	12.7
					1	1	6.37
					2	1.5	4.24
					4	2.5	2.55
					6	3.5	1.8

The **second stage $TF_2(p)$** has the same topology than the fast shaper but the approximation is different due to other dominant parameters:

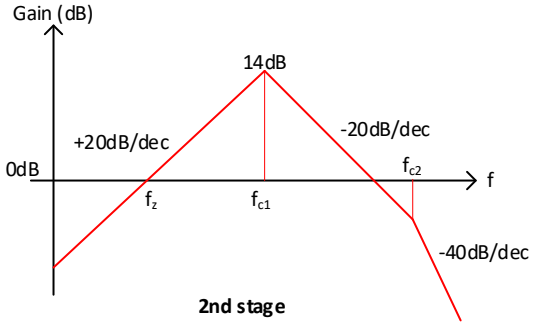
$$TF_{2SS}(p) \cong \frac{R_f.C_{IN}.p}{(1+\tau.p)^2.(1+\frac{p}{A_0.w_0})}, f_{c1} = \frac{1}{2.\pi.\tau} = \frac{1}{2.\pi.RC}, f_z = \frac{1}{2.\pi\ R_f.C_{IN}}, G_{fc2} \cong \frac{R_f}{2.R_{IN}}, f_{c2} = \frac{A_0.w_0}{2.\pi} = 251MHz$$

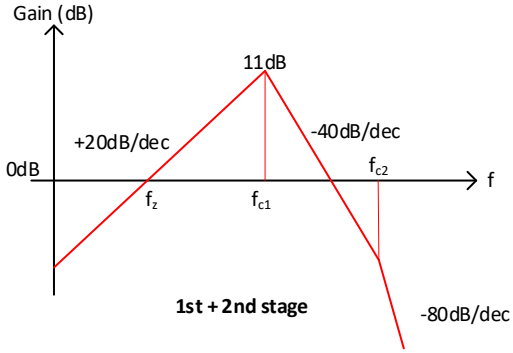
Parameters		Full equation		Approximation	
Ri (Ω)	10k	$2.\tau + \frac{1}{A_0.w_0} + \frac{R_f.C_{IN}}{A_0.}$	2.59E-8 1.27E-7	$2\tau + \frac{1}{A_0.w_0}$	2.56E-8 1.26E-7
Ci (pF) **Variable**	1.25 6.25	$\tau^2 + \frac{2.\tau}{A_0.{}^2w_0} + \frac{R_f.C_{IN}}{A_0.w_0}$	2.51E-16 4.38E-15	$\tau^2 + \frac{2.\tau}{A_0.w_0}$	1.72E-16 3.99E-15
Rf (kΩ)	100	$\frac{\tau^2}{A_0.w_0}$	9.92E-26 2.48E-24	$\frac{\tau^2}{A_0.w_0}$	9.92E-26 2.48E-24
Cf (pF) **Variable**	0.125 0.625	$R_f.C_{IN}$	1.25E-7 6.28E-7	f_p (MHz)	12.7 2.55
A0	398			f_z (MHz)	1.27 0.255
f0 (KHz)	630			G_{fc}	5

Slow shaper full stage: $TF_{SS}(p) = TF_1(p).TF_2(p) \cong \frac{R_f.C_{IN}.p}{(1+\tau.p)^3.(1+\frac{p}{A_0.w_0})^2}$

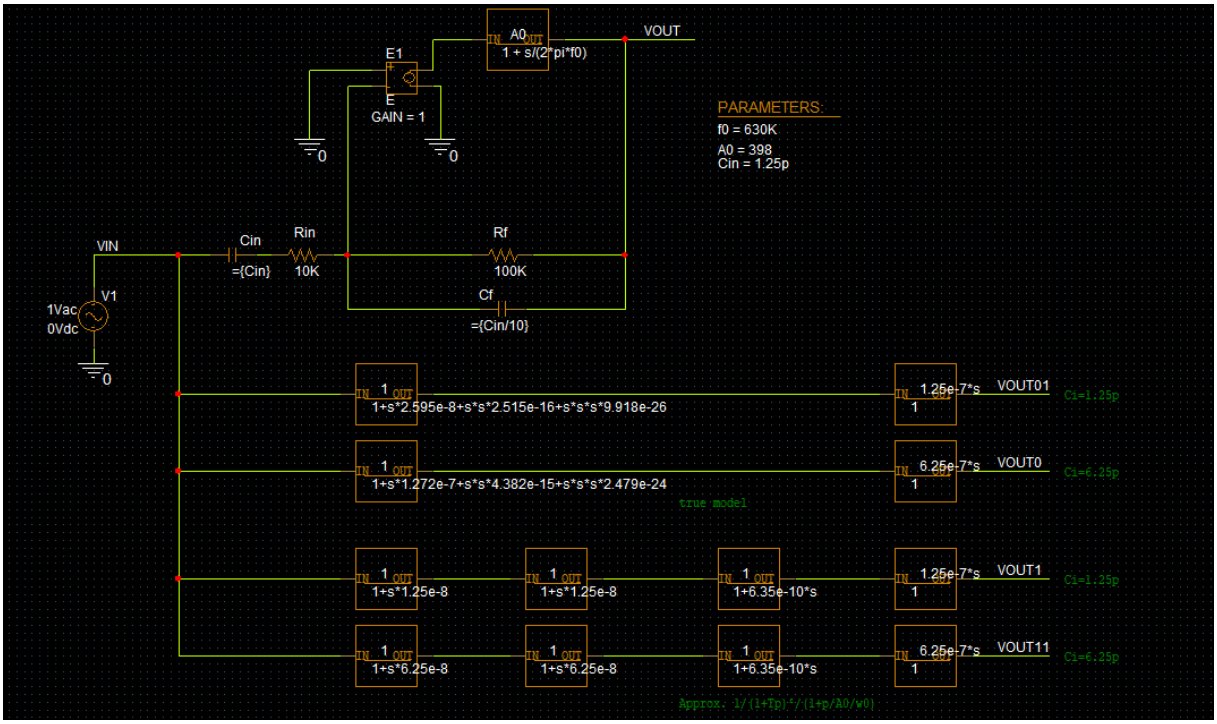
$$f_{c1} = \frac{1}{2.\pi.\tau} = \frac{1}{2.\pi.RC}, f_z = \frac{1}{2.\pi\ R_f.C_{IN}}, G_{fc} \cong \frac{R_f}{2.\sqrt{2}.R_{IN}}, f_{c2} = \frac{A_0.w_0}{2.\pi} = 251MHz$$

OPA	A_0 (dB)	A_0	f_0 (KHz)	S/C	C_{IN} (pF)	C_f (pF)	G_{fc2} (dB)	G_{fc} (dB)	f_{c1} (MHz)	f_z (MHz)	f_{c2} (MHz)
SS	52	398	630	0	1.25	0.125	14	11	12.7	1.27	251
				1	2.5	0.25			6.37	0637	
				2	3.75	0.375			4.24	0.424	
				4	6.25	0.625			2.55	0.255	
				6	8.75	0.875			1.8	0.18	

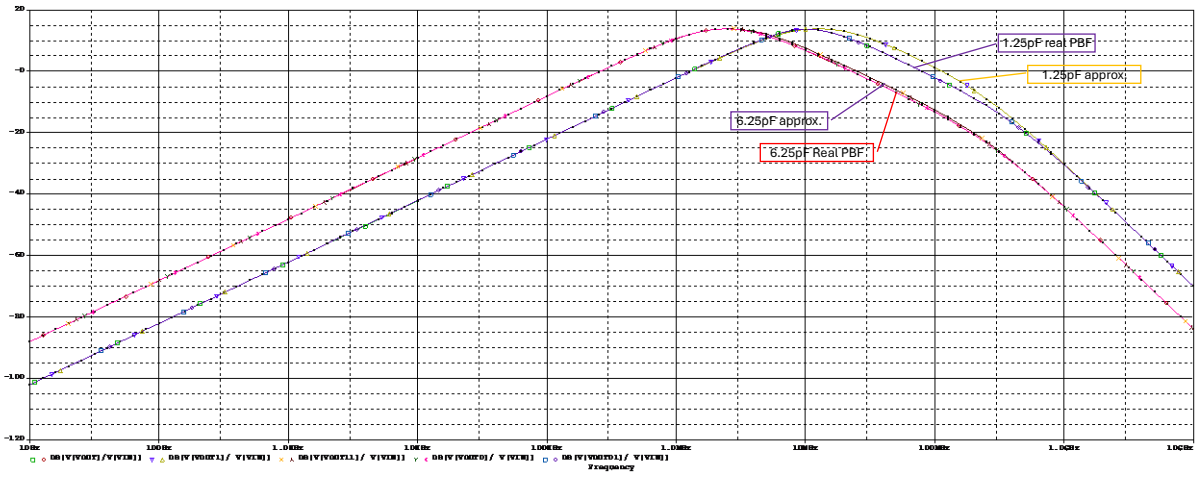




5.1 1st stage model and approximation AC simulation



Cin(F) = 1.25p 6.25p



The 1st stage approximation model is accurate for 6.25pF, and for 1.25pF, it is accurate except between f_{c1} and f_{c2} where it as max. +3dB difference.

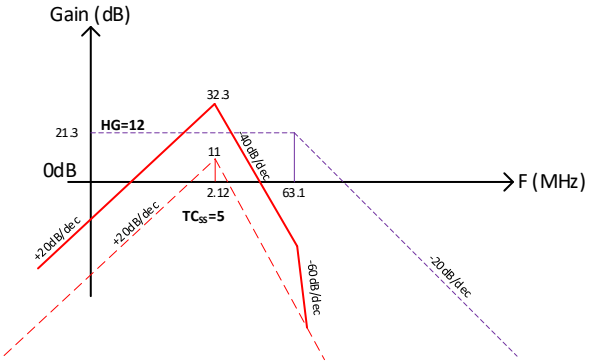
6 PREAMPLIFER + SLOW SHAPER

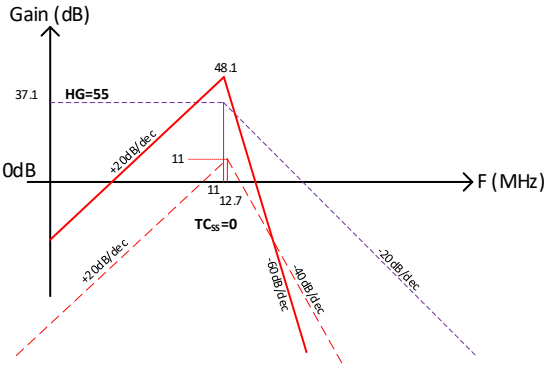
Full stage: $TF_{PASH}(p) = TF_{PA}(p).TF_{SS}(p) \cong \frac{R_f.C_{IN}.p}{(1+\tau_{SS}.p)^3.\left(1+\frac{p}{A_{0ss}.w_{0ss}}\right)^2\left(1+\frac{p}{A_{0pa}.w_{0pa}}\right)}$

Preamplifier stage: $f_{cpa} = \frac{C_{fpa}.A_{0pa}.w_{0pa}}{2.\pi.(C_{fpa}+C_{ipa})}$ (depends on S/C HG set)

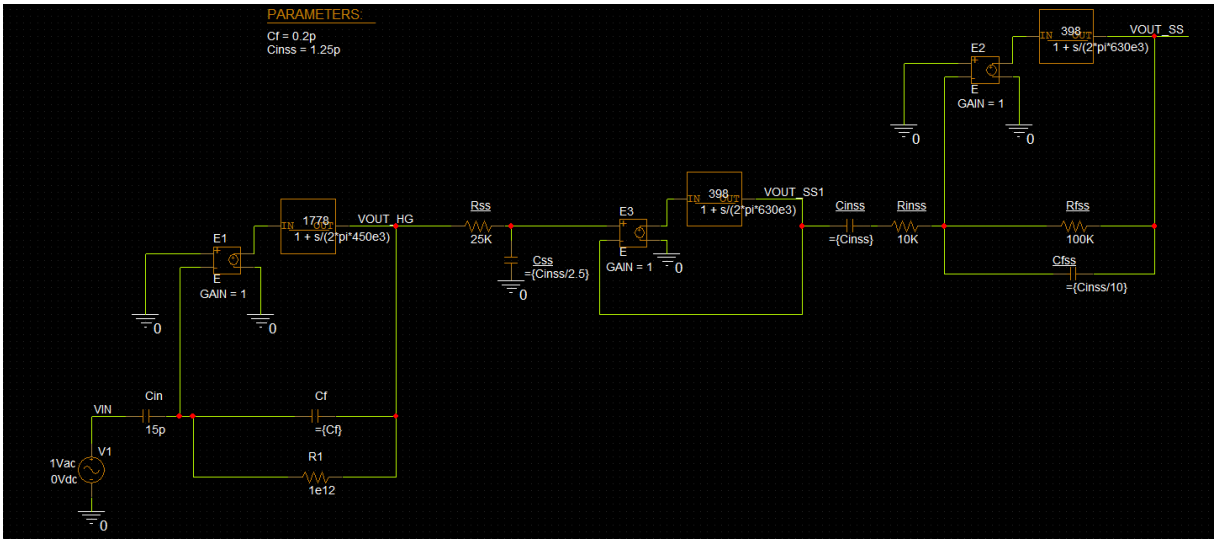
Slow shaper stage: $f_{css} = \frac{1}{2.\pi.\tau_{SS}} = \frac{1}{2.\pi.R_{INSS}C_{INSS}}$ (depends on S/C TCss set), $G_{fcss} \cong \frac{R_f}{2.\sqrt{2}.R_{IN}}$, $f_{c2} = \frac{A_{0ss}.w_{0ss}}{2.\pi}$ = 251MHz

OPA	S/C HG gain	C_f (pF)	G_{0pa} (dB)	f_{cpa} (MHz)	OPA	S/C TC_{SS}	C_{INSS} (pF)	G_{fcss} (dB)	f_{css} (MHz)	f_{c2} (MHz)
HG **A_0=65dB** **F_0=450K**	62	0.025	53.0	1.8	**SS** **A_0=52dB** **F_0=630K**	**0**	**1.25**	11	12.7	251
	55	0.2	**37.1**	**11.0**		1	2.5		6.37	
	48	0.375	31.8	20.0		2	3.75		4.24	
	31	0.8	25.4	41.0		3	5		3.18	
	20	1.075	22.8	54.0		4	6.25		2.55	
	12	1.275	**21.3**	**63.1**		**5**	**7.5**		**2.12**	
	0	1.575	19.5	76.5		6	8.75		1.8	



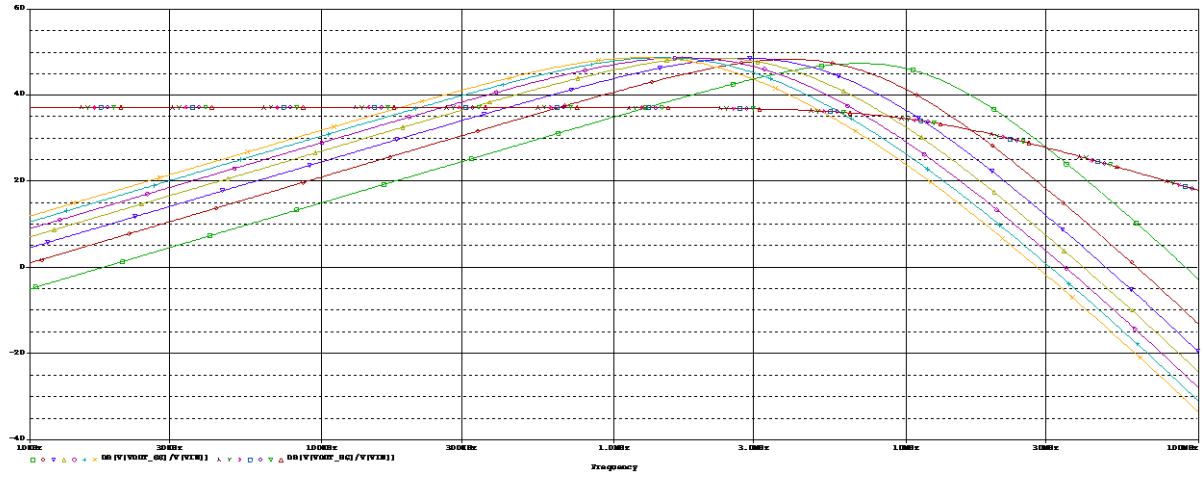


6.1 AC simulation



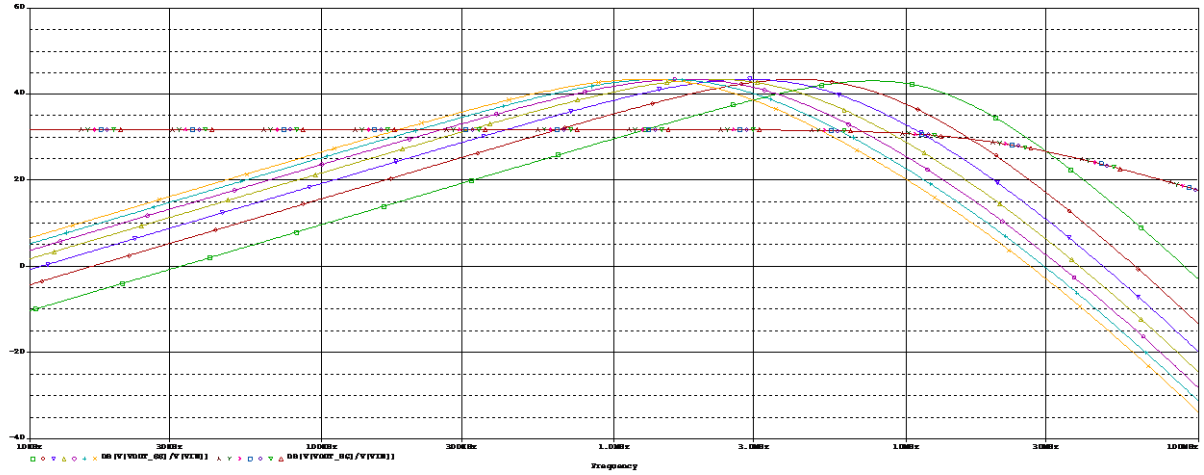
HG=55 (f_{cpa}=11MHz)

C_{INSS}(F)= 1.25p 2.5p 3.75p 5p 6.25p 7.5p 8.75p



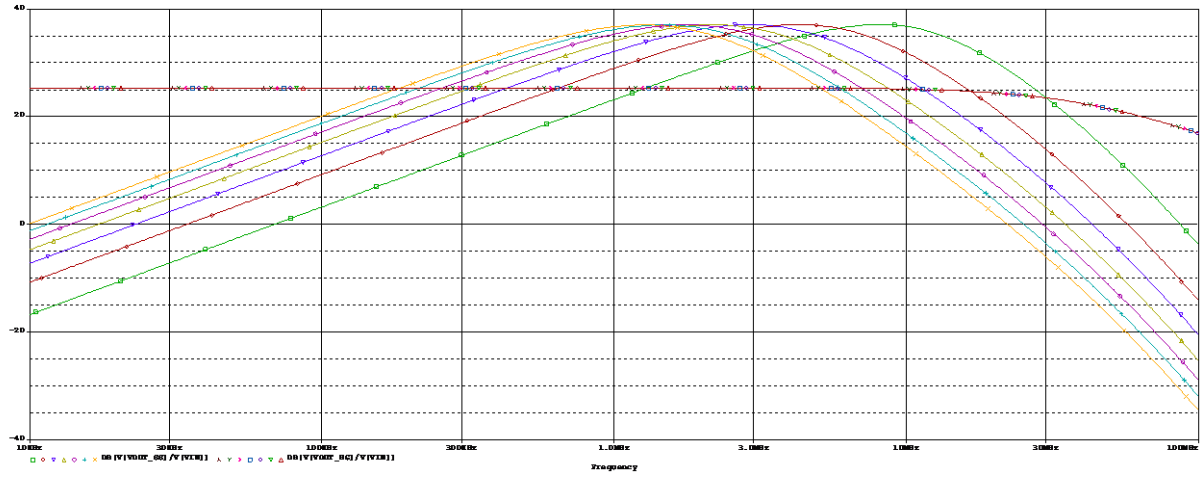
HG=48 (f_{cpa}=20MHz)

C_{INSS}(F)= 1.25p 2.5p 3.75p 5p 6.25p 7.5p 8.75p



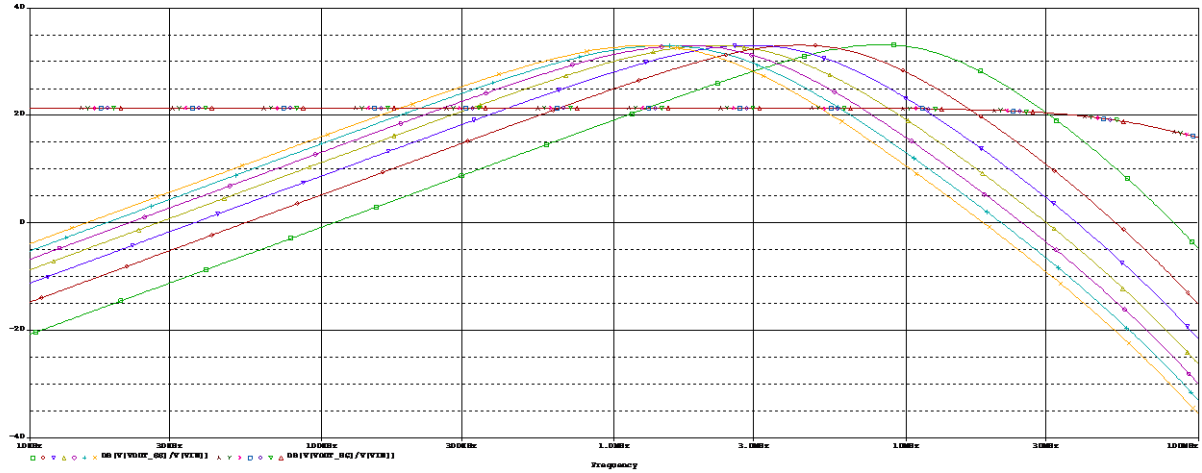
HG=31 (f_{cpa}=41MHz)

C_{INSS}(F)= 1.25p 2.5p 3.75p 5p 6.25p 7.5p 8.75p



HG=12 (f_{cpa}=63MHz)

C_{INSS}(F)= 1.25p 2.5p 3.75p 5p 6.25p 7.5p 8.75p

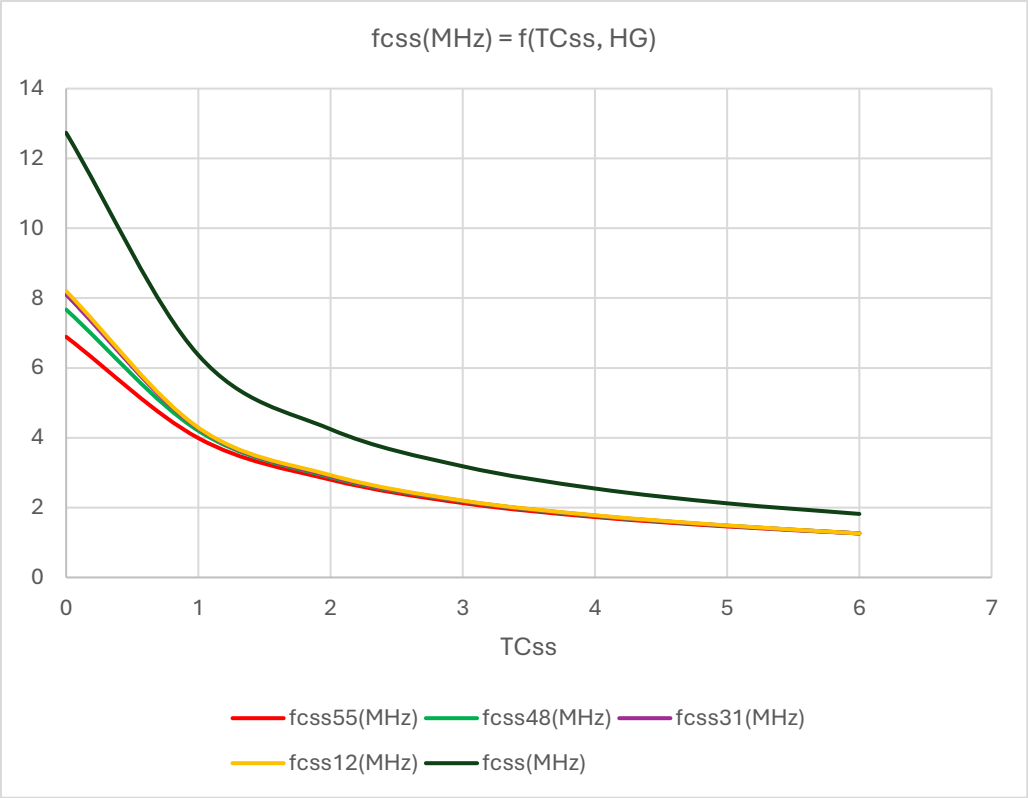


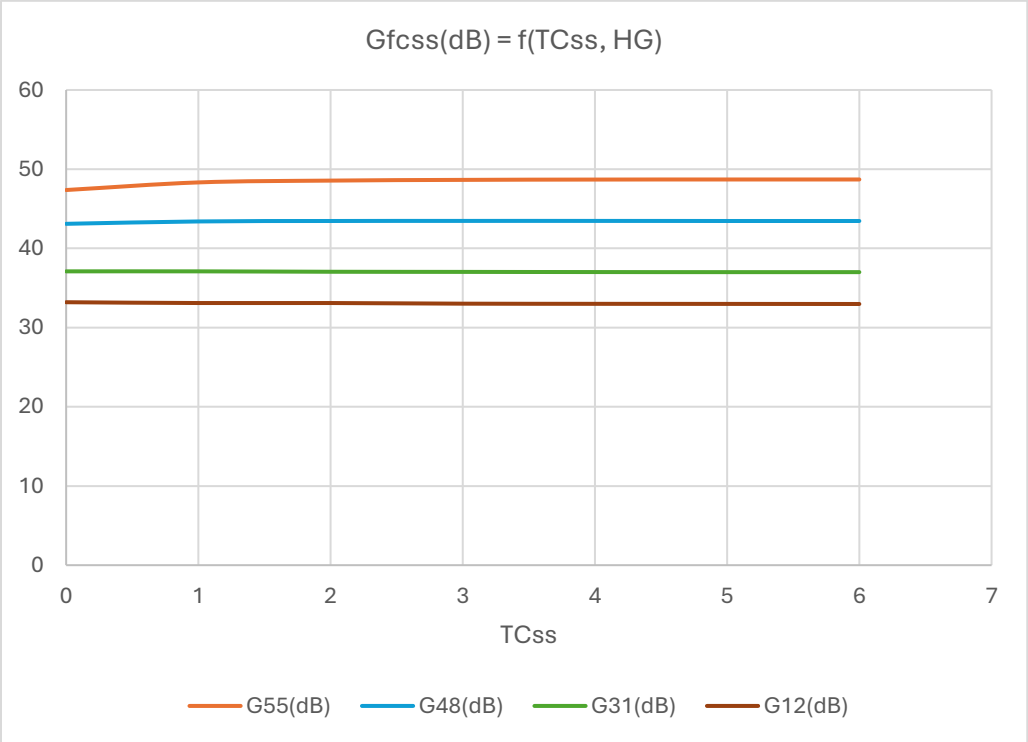
		HG=55		HG=48		HG=31		HG=12		Theory	
Cinss (pF)	TC	fcss55 (MHz)	G55 (dB)	fcss48 (MHz)	G48 (dB)	fcss31 (MHz)	G31(dB)	fcss12 (MHz)	G12(dB)	f_{css} (MHz)	f_{css}/f_{css12}
1.25	0	6.89	47.37	7.67	43.1	8.1	37.1	8.2	33.2	12.73	1.55
2.5	1	3.99	48.33	4.2	43.4	4.27	37.1	4.29	33.1	6.37	1.48
3.75	2	2.8	48.56	2.87	43.46	2.9	37.05	2.93	33.1	4.24	1.45
5	3	2.13	48.65	2.18	43.47	2.19	37.03	2.2	33.02	3.18	1.45
6.25	4	1.73	48.69	1.76	43.47	1.77	37.01	1.78	33	2.55	1.43
7.5	5	1.46	48.7	1.48	43.46	1.48	37	1.49	32.99	2.12	1.42
8.75	6	1.26	48.7	1.26	43.46	1.26	37	1.26	32.98	1.82	1.44

The f_{css} frequency shift observed on simulation versus theory is due to the +20dB/dec => -40dB/dec and the ratio between theory and simulation is ~1.4/1.5.

When the HG is high, the -20dB/dec LPF add another shift (+20dB/dec => -60dB/dec).

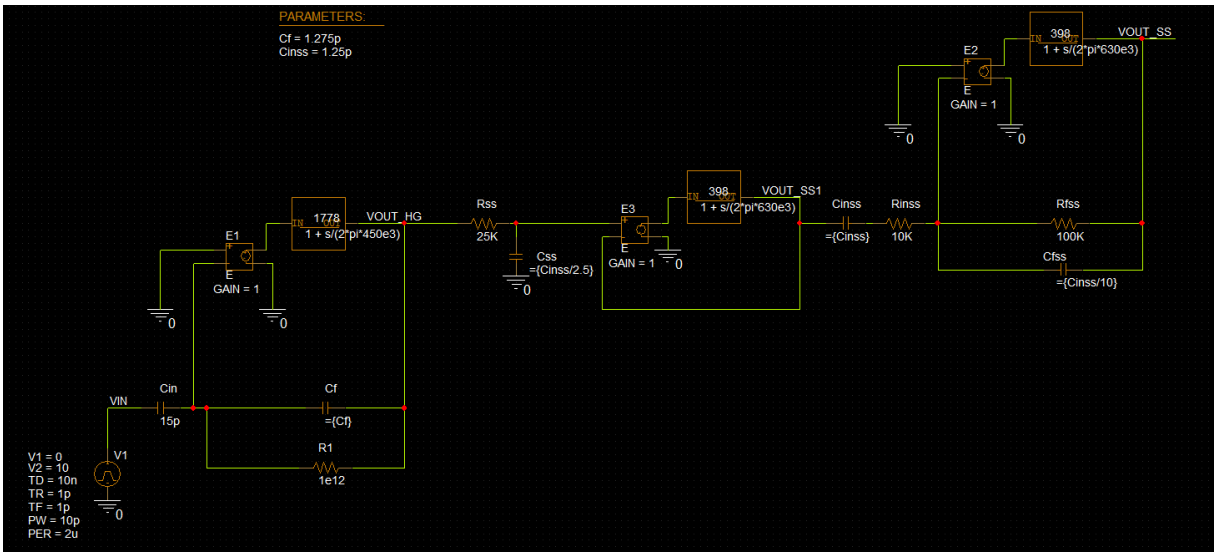
TODO: Ratio to find theoretically


fcss(MHz) = f(TCss, HG)
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
TCss
fcss55(MHz)
fcss48(MHz)
fcss31(MHz)
fcss12(MHz)
fcss(MHz)


Gfcss(dB) = f(TCss, HG)
60
50
40
30
20
10
0
0
1
2
3
4
5
6
7
TCss
G55(dB)
G48(dB)
G31(dB)
G12(dB)

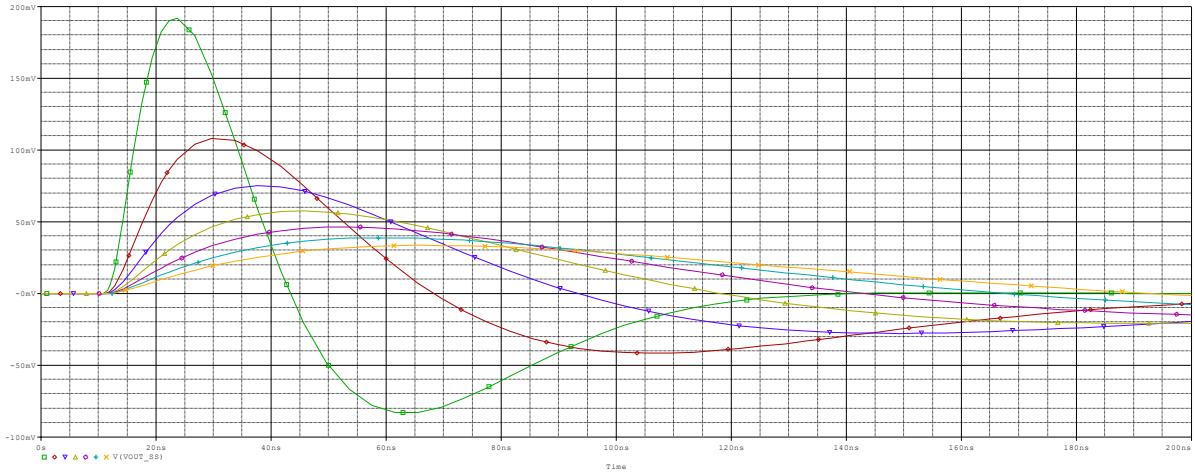
6.2 Transient simulation

6.2.1 Dirac input

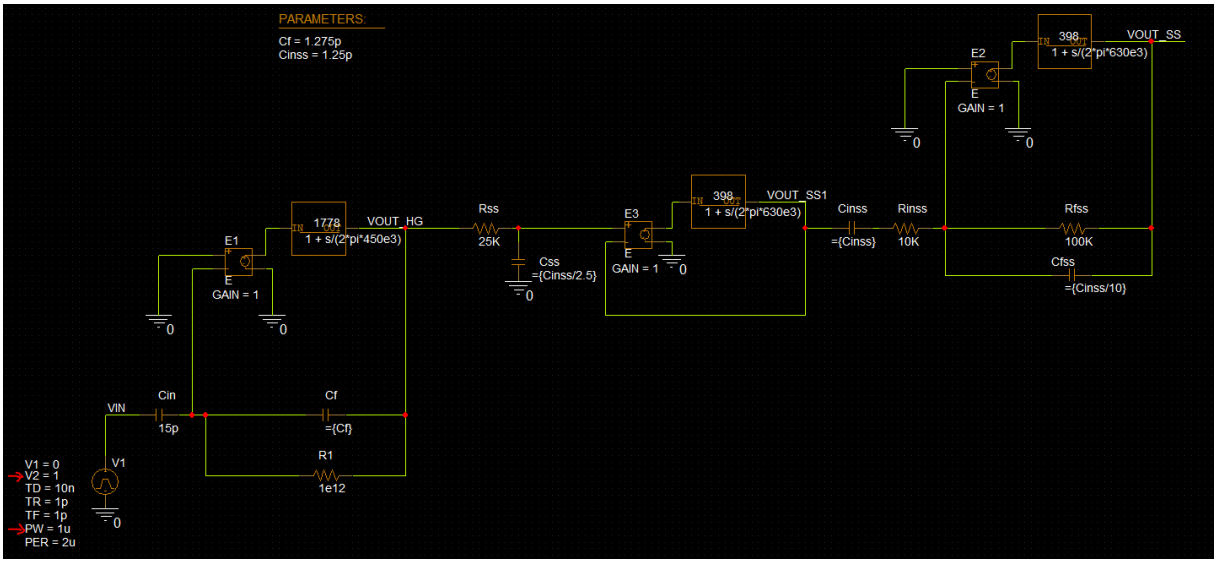


HG=12 (f_{cpa}=63MHz)

C_{INSS}(F)= 1.25p 2.5p 3.75p 5p 6.25p 7.5p 8.75p



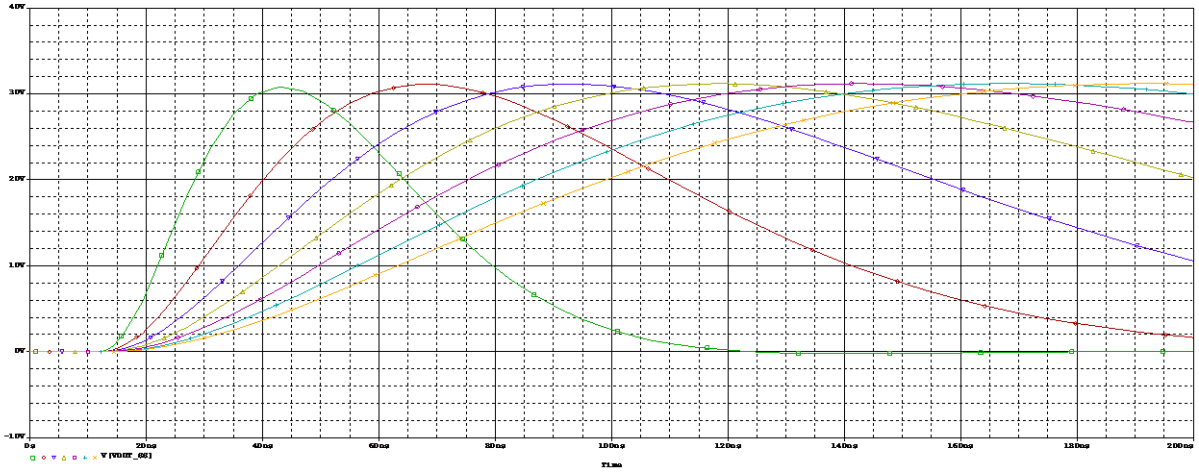
6.2.2 Step input



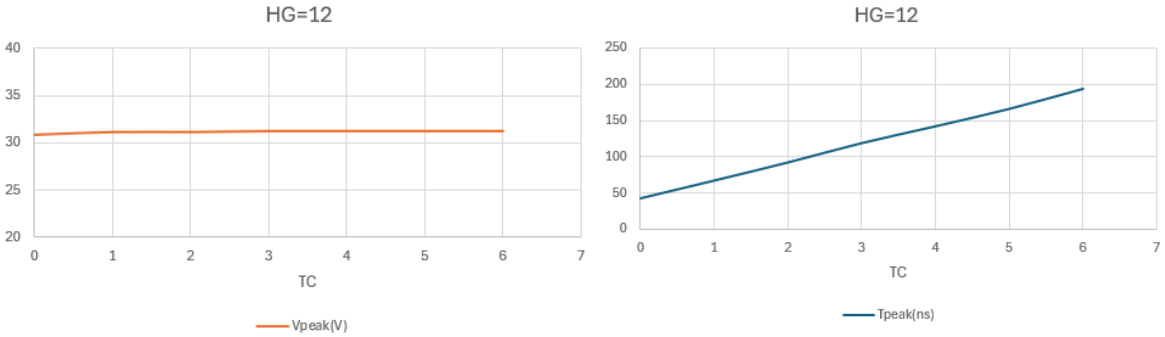
Varying TC for several HG=12:

HG=12 (f_{cpa}=63MHz)

C_{INSS}(F)= 1.25p 2.5p 3.75p 5p 6.25p 7.5p 8.75p



		HG=12	
Cinss(pF)	**TC**	**Tpeak(ns)**	**Vpeak(V)**
1.25	0	43.3	30.8
2.5	1	67.8	31.1
3.75	2	92.8	31.1
5	3	119.4	31.2
6.25	4	142.6	31.2
7.5	5	166.6	31.2
8.75	6	194.2	31.2



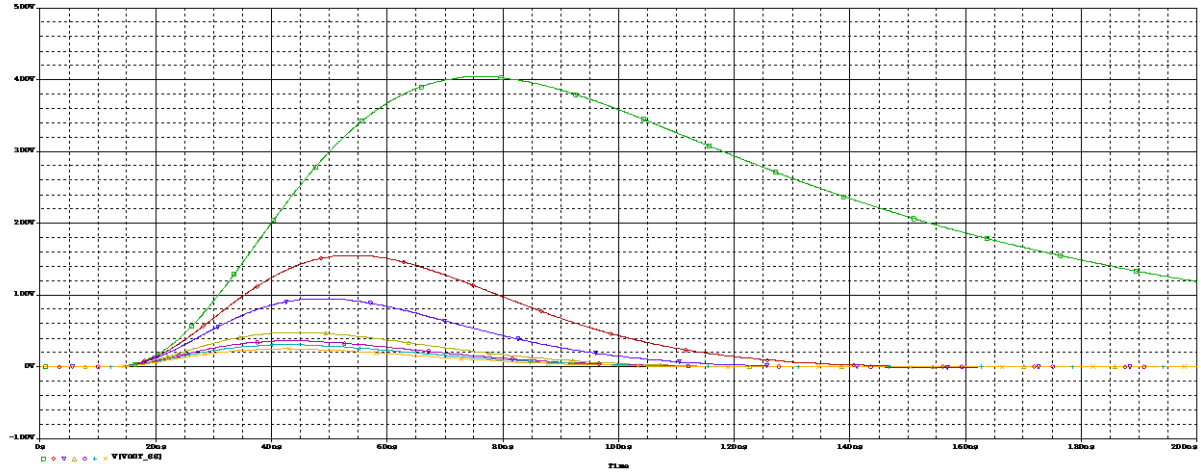
⇨ **For a dedicated HG, Peaking Time is linear with TC and gain is constant.**

<u>Varying HG for several TC:</u>

TC=0

C_f(F)= 0.025p 0.2p 0.375p 0.8p 1.075p 1.275p 1.575p

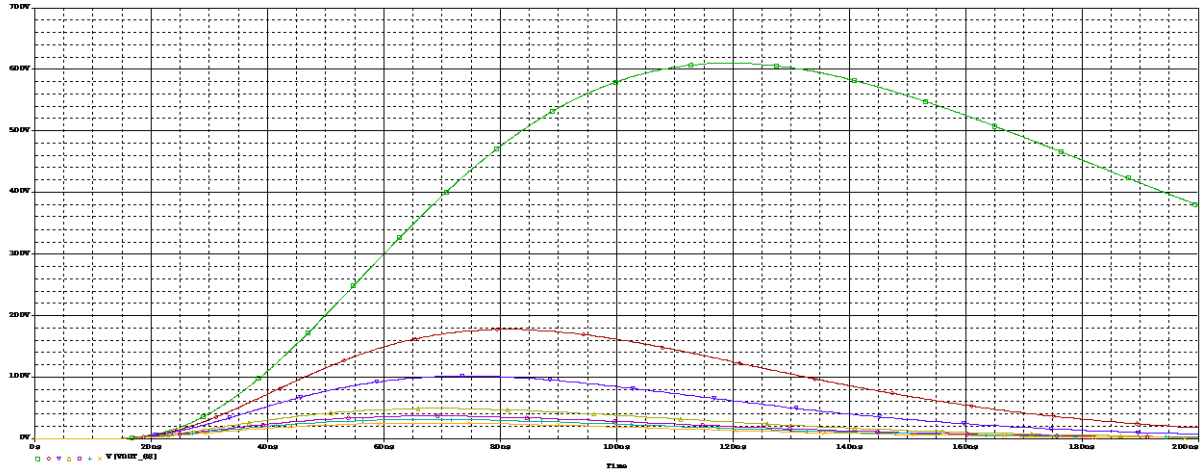
HG= 62 55 48 31 20 12 0



TC=1

C_f(F)= 0.025p 0.2p 0.375p 0.8p 1.075p 1.275p 1.575p

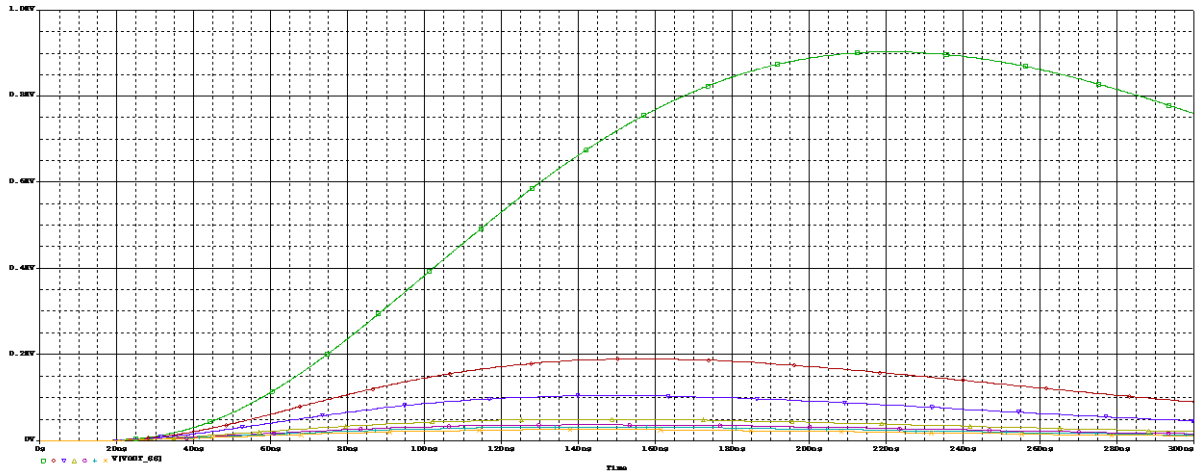
HG= 62 55 48 31 20 12 0



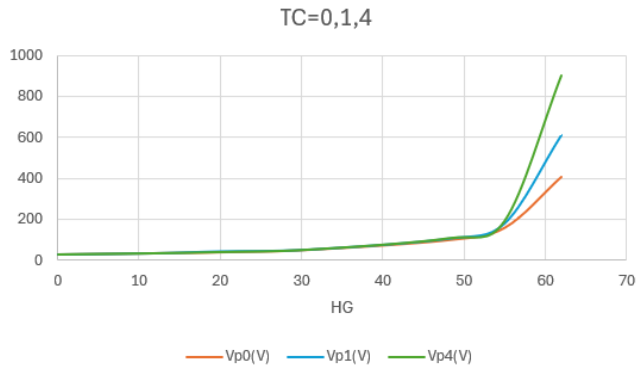
TC=4

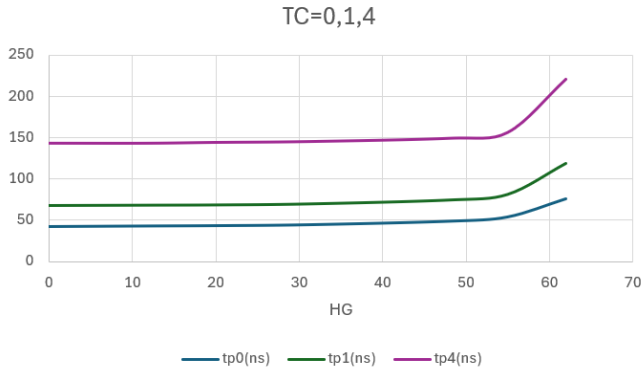
C_f(F)= 0.025p 0.2p 0.375p 0.8p 1.075p 1.275p 1.575p

HG= 62 55 48 31 20 12 0



		TC=0 (1.25pF)		TC=1 (2.5pF)		TC=4 (6.25pF)	
Cf(pF)	HG	tp0(ns)	Vp0(V)	tp1(ns)	Vp1(V)	tp4(ns)	Vp4(V)
0.025	62	76.2	404	119	610	221	902
0.2	55	54	155	81.1	177	156	189
0.375	48	48.8	95	74.3	102	149	104
0.8	31	44.5	48	69.5	49	145	49
1.075	20	43.4	36	68.3	40	144	37
1.275	12	43	31	68	31	143	31
1.575	0	42.3	25	67.6	25	143	25





⇨ **For a dedicated TC, Peaking Time is quite constant up to HG=55 where the influence of the pre-amplifier cutoff frequency begins to play a role since it is overlapps with the BPF centered frequency.**